

NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026

Frontend Web Development using React

Project Submission Template

HR EMPLOYEE MANAGEMENT TOOL PROJECT

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TITLE : HR EMPLOYEE MANAGEMENT TOOL PROJECT

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1. ABSTRACT

The HR Leave Management Tool is a web-based application designed to digitize and automate the employee leave request and approval process within an organization. It provides a centralized platform where employees can apply for leave, view their leave balance, and track request status, while HR managers and team leads can approve, reject, or monitor leave requests in real-time. Built with React for the frontend and a REST API backend, the tool reduces manual paperwork, eliminates communication gaps, and ensures transparency and compliance with company leave policies. The system improves operational efficiency and saves HR time by automating notifications, leave balance calculations, and report generation.

2. PROBLEM STATEMENT

Most small to mid-sized companies still manage employee leave using spreadsheets, emails, or paper forms. This manual process leads to delays in approvals, loss of leave records, errors in leave balance calculations, lack of transparency, and difficulty in generating reports for payroll and compliance. Employees often do not know their remaining leave balance, and managers struggle to track overlapping leaves. There is a need for a simple, centralized, and user-friendly system that automates the entire leave lifecycle.

3. OBJECTIVES

1. Automate leave application, approval, and rejection workflows.
2. Provide real-time visibility of leave balance and leave history to employees.
3. Enable managers and HR to approve/reject requests and view team calendars.
4. Prevent leave conflicts by showing overlapping absences.
5. Generate reports for HR and payroll use.
6. Reduce manual effort and paperwork for HR department by 80%.
7. Ensure data security and role-based access for employees, managers, and HR.

4. PROJECT SCOPE

Included Features:

1. Employee login/signup with role-based access: Employee, Manager, HR Admin.
2. Apply for leave with leave type, date range, reason, and attachment upload.
3. View leave balance for Casual, Sick, Earned, and Unpaid Leave.
4. Track leave request status: Pending, Approved, Rejected.
5. Manager dashboard to approve/reject requests and view team calendar.
6. HR dashboard for leave policy setup, user management, and reports export to Excel/PDF.
7. Email and in-app notifications for status updates.
8. Holiday calendar management.

Limitations / Out of Scope:

1. Payroll integration is not included in v1.0.
2. Mobile app is not developed; system is responsive web only.
3. Advanced analytics and AI-based leave prediction are future enhancements.
4. Integration with biometric attendance systems planned for future phase.

5. TECHNOLOGIES USED

1. Frontend: React 18, React Router, Axios, Tailwind CSS, React Hook Form, React Toastify, <http://Chart.js>.
2. State Management: React Context API + useReducer for global state.
3. Backend: <http://Node.js>, <http://Express.js>, REST API.
4. Database: MongoDB with Mongoose ODM.
5. Authentication: JWT for token-based auth, bcrypt for password hashing.
6. File Storage: Cloudinary for leave document uploads.
7. Email: Nodemailer with SMTP.

8. ***Deployment***: Frontend on Vercel, Backend on Render, Database on MongoDB Atlas.
9. ***Tools***: VS Code, Git, GitHub, Postman for API testing.

6. APPLICATION FLOW

1. ***Login***: User logs in with email/password. JWT token is generated and stored.
2. ***Dashboard***: Employee sees leave balance, upcoming leaves, and quick “Apply Leave” button.
3. ***Apply Leave***: Employee selects leave type, start/end date, reason, uploads document if required, and submits. System validates dates and balance.
4. ***Notification***: Manager receives email/in-app notification.
5. ***Approval***: Manager reviews request on dashboard, sees team calendar to check conflicts, then approves or rejects with comment.
6. ***Update***: Employee gets notified. Leave balance updates automatically if approved.
7. ***HR View***: HR can override requests, manage users, set leave policies, and export monthly reports.
8. ***Calendar View***: All users can view company-wide approved leaves in calendar format to plan work.

7. KEY REACT CONCEPTS USED

1. ***Functional Components & Hooks***: Used ``useState``, ``useEffect``, ``useContext`` for state and side effects instead of class components for cleaner code.
2. ***React Router***: Implemented client-side routing for ``/login``, ``/dashboard``, ``/apply-leave``, ``/admin`` with protected routes using ``Navigate``.
3. ***Context API***: Managed global auth state and user role to avoid prop drilling across components.
4. ***Controlled Forms***: Used React Hook Form for form validation and performance optimization on leave application form.

5. ***Conditional Rendering***: Showed different UI for Employee vs HR role based on role in context.
6. ***Axios Interceptors***: Attached JWT token to every API request and handled 401 errors globally.
7. ***Custom Hooks***: Created `useAuth` and `useLeaveBalance` hooks for reusable logic.
8. ***Component Composition***: Built reusable components like `LeaveCard`, `StatusBadge`, `DataTable` to maintain DRY principle.

8. EXPECTED OUTPUT

Final Outcome: A fully functional web app where employees can apply for leave in under 2 minutes, managers can approve/reject in one click, and HR gets accurate reports instantly.

Key Pages/Features

1. ***Login Page***: Clean UI with role-based redirect.
2. ***Employee Dashboard***: Cards showing leave balance, recent requests, and a monthly calendar heatmap.
3. ***Apply Leave Form***: Date picker with disabled weekends/holidays, real-time balance check.
4. ***Manager Dashboard***: Table of pending requests with Approve/Reject buttons and team calendar view.
5. ***HR Reports Page***: Filter by date, department, leave type. Export to CSV/PDF.

9. CHALLENGES FACED

1. ***Date Handling & Timezones***: Leave dates were mismatched due to timezone issues. Resolved by storing all dates in UTC and using `date-fns` for formatting.
2. ***Overlapping Leave Validation***: Preventing double booking required complex date range logic. Solved using interval overlap checks on backend before saving.
3. ***State Sync After Approval***: UI didn't update after manager action. Fixed by refetching data and updating Context state post API call.
4. ***File Upload Security***: Risk of malicious file uploads. Resolved by restricting file types, size limit, and using Cloudinary for virus scanning.
5. ***Role-Based Routing***: Preventing employees from accessing HR routes. Implemented route guards and backend middleware checks.

10. LEARNING OUTCOMES

1. Gained hands-on experience with full-stack development using MERN stack.
2. Understood JWT authentication, role-based access control, and secure API design.
3. Learned advanced React patterns: custom hooks, Context API, form optimization.
4. Improved skills in database schema design and MongoDB aggregation for reports.
5. Practiced Git workflow, deployment pipelines, and debugging production issues.
6. Learned to write clean, maintainable code and document APIs using Postman.

11. CONCLUSION

The HR Leave Management Tool successfully replaces manual leave tracking with an automated, transparent, and efficient system. It solves the core problems of delayed approvals, inaccurate leave balances, and poor visibility by providing role-based dashboards, real-time notifications, and reporting. Through this project, the leave process time was reduced from 2-3 days to under 30 minutes, and HR workload decreased significantly. The application is scalable and can be extended with payroll integration and mobile support in future versions. It demonstrates how modern web technologies can streamline HR operations and improve employee experience.